

1, 12, and so on. We can write the beginning of this continued fraction in a more conventional style as

$$\text{mysterynumber} = 2 + \frac{1}{1 + \frac{1}{2 + \frac{1}{1 + \frac{1}{1 + \frac{1}{4 + \frac{1}{1 + \dots}}}}}}$$

In fact, this mystery number is none other than our old friend e , the base of the natural logarithm! The constant e , just like ϕ and other irrational numbers, has an infinite decimal expansion with no apparent pattern and cannot be represented by a finite fraction, and it seems like it's impossible to express its exact numeric value concisely. But by using the new concept of continued fractions and a new concise notation, we can write these apparently intractable numbers in one line. There are also several remarkable ways to use continued fractions to represent π . This is a victory for data compression. It's also a victory in the perennial battle between order and chaos: where we thought there was nothing but encroaching chaos dominating the numbers we love, we find that there was always a deep order beneath the surface.

Our continued fraction for ϕ came from a special equation that works only for ϕ . But in fact, it is possible to generate a continued fraction representation of any number.

An Algorithm for Generating Continued Fractions

To find a continued fraction expansion for any number, we'll use an algorithm.

It's easiest to find continued fraction expansions for numbers that are integer fractions already. For example, consider the task of finding a continued fraction representation of $105/33$. Our goal is to express this number in a form that looks like the following: